

4.6.2 Triple Integrals in Spherical Coordinates_Guide

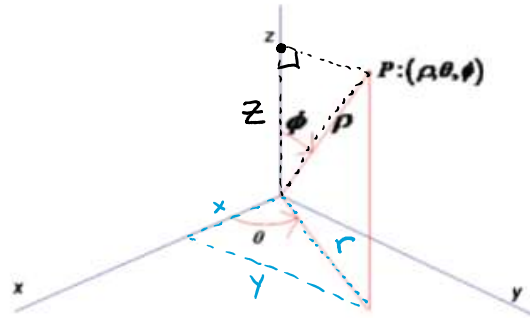
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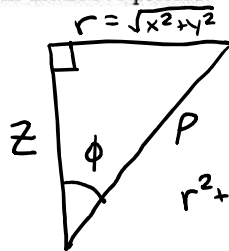
4.6.2 Triple
Integrals i...

4.6.2 Triple Integrals in Spherical Coordinates

Like cylindrical coordinates, the spherical coordinate system is used to make certain triple integrals easy to evaluate. A picture of a point given by its spherical coordinates is:



When trying to derive equations that relate rectangular and spherical coordinates we obtain the following right triangle from the above picture:



$$r^2 + z^2 = x^2 + y^2 + z^2 = \rho^2$$

$$\text{Note } \cos(\phi) = \frac{z}{\rho} \rightarrow z = \rho \cos(\phi)$$

$$\sin(\phi) = \frac{r}{\rho} \rightarrow r = \rho \sin(\phi)$$

$$\text{Also } x = r \cos(\theta) \text{ \& } y = r \sin(\theta)$$

To Convert from Spherical to Rectangular Coordinates:

Use the following equations: $x = \rho \sin(\phi) \cos(\theta)$, $y = \rho \sin(\phi) \sin(\theta)$, and $z = \rho \cos(\phi)$.

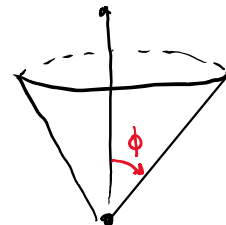
Remark: We restrict ρ so that $\rho \geq 0$, and we restrict ϕ so that $0 \leq \phi \leq \pi$. This means that

$$\rho = \sqrt{x^2 + y^2 + z^2} \Leftrightarrow \rho^2 = x^2 + y^2 + z^2.$$

Example 1: Describe the graph of the following equation in spherical: $\phi = \frac{\pi}{6}$.

Note that ρ & θ are not restricted by this equation & therefore can take on any value.

A cone!



Fact: Suppose that $E = \{(\rho, \theta, \phi) : \alpha \leq \theta \leq \beta, c \leq \phi \leq d, g_1(\theta, \phi) \leq \rho \leq g_2(\theta, \phi)\}$ where $\beta - \alpha \leq 2\pi$, $d - c \leq \pi$, and both $g_1(\theta, \phi)$ and $g_2(\theta, \phi)$ are continuous and nonnegative for all possible values of θ and ϕ . Then, assuming that $f(x, y, z)$ is continuous, we have that

$$\iiint_E f(x, y, z) dV = \int_{\alpha}^{\beta} \int_c^d \int_{g_1(\theta, \phi)}^{g_2(\theta, \phi)} f(x, y, z) \rho^2 \sin(\phi) d\rho d\phi d\theta.$$

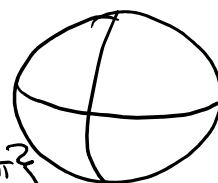
Note: The above formula is partially justified on pages 1051 and 1052 of the textbook.

Note: Since theta and phi are bounded by constants their order can be reversed.

Note: The general procedure for setting up triple integrals as iterated integrals does not extend in a natural way to spherical since, currently; it doesn't make sense to project onto the $\phi\theta$ -plane (this will be revisited in section 4.7). We will therefore take a more intuitive approach.

Example 2: Evaluate $\iiint_B e^{(x^2+y^2+z^2)^{1/2}} dV$ where $B = \{(x, y, z) : x^2 + y^2 + z^2 \leq 1\}$.

We note that B is a sphere of radius 1
w. th all interior points included.



$$B = \{(\rho, \phi, \theta) \mid 0 \leq \rho \leq 1, 0 \leq \theta \leq 2\pi, 0 \leq \phi \leq \pi\}$$

$$\iiint_B e^{\sqrt{x^2+y^2+z^2}} dV = \int_0^{2\pi} \int_0^{\pi} \int_0^1 e^{\rho} \rho^2 \sin(\phi) d\rho d\phi d\theta$$

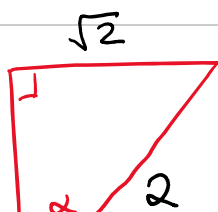
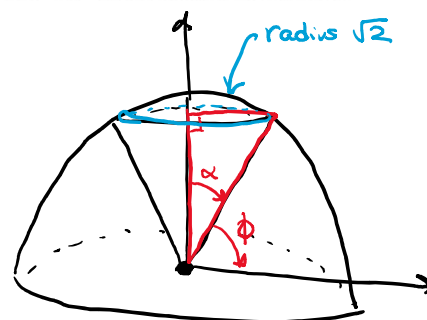
Example 3: Find the exact volume of the solid that lies inside the sphere $x^2 + y^2 + z^2 = 4$, above the xy -plane, and below the half-cone $z = \sqrt{x^2 + y^2}$.

We first sketch E .

Clearly $0 \leq \theta \leq 2\pi$

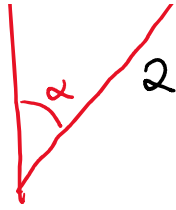
& $0 \leq \rho \leq 2$

The bounds on ϕ are harder



Note the hypotenuse of this triangle is 2 since it is the radius of the sphere.

Thus, the side of the triangle is the radius of the sphere.



is the radius of the sphere.

The top side of the triangle is the radius of Circle (cone) when the sphere & the cone intersect.

The cone & sphere intersect when

$$z = \sqrt{x^2 + y^2} \quad \& \quad x^2 + y^2 + z^2 = 4$$

$$x^2 + y^2 + (\sqrt{x^2 + y^2})^2 = 4 \rightarrow 2x^2 + 2y^2 = 4 \rightarrow x^2 + y^2 = 2$$

So the top of the triangle has length $\sqrt{2}$.

$$\text{So } \sin(\alpha) = \frac{\sqrt{2}}{2} \rightarrow \alpha = \frac{\pi}{4}$$

$$\text{So } \frac{\pi}{4} \leq \phi \leq \frac{\pi}{2}$$

$$V = \iiint_E 1 \, dV = \int_0^2 \int_{\pi/4}^{\pi/2} \int_0^{2\pi} \rho^2 \sin(\phi) \, d\theta \, d\phi \, d\rho$$